LESSON PLAN

BASIC RADIO & AVIONICS (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principle of operation of various engine instruments, viz, torque meter, thrust meter, oil pressure gauges, fuel flow meter, jet pipe temper
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principle of operation of various engine instruments, viz, torque meter, thrust meter, oil pressure gauges, fuel flow meter, jet pipe temperature indicator, throttle lever, Position indicator, tachometer etc. Introduction to concept of glass cockpit
5. Specific Objectives	25.1.2.1 Trainee will be able to know and understand the constructional details of starters, ignition units and principles of. operation of various engine instruments

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of instrument panel layout and identification of engine instruments; Instructor supervision required for safety
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary instruments located on the engine instrument panel?
(b) Define the function of a tachometer in engine operation.
(c) List three safety devices associated with the electrical system.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Tachometer principle and construction • Torque meter operating mechanism • Oil pressure gauge sensing system • Fuel flow meter measurement technique • Jet pipe temperature indicator function	L & Ds	PPT, IB	15 min
(b)	• Throttle lever and position indicator operation • Glass cockpit integration and display • Interconnection of engine instruments • Instrument calibration and accuracy factors • Troubleshooting common instrument failures	L & Ds	PPT, PDF	20 min
(c)	• Ignition unit principles and construction • Starter system operation basics • Engine instrument warning systems • Maintenance requirements for instruments • Future trends in engine monitoring	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the torque meter measures engine output.
(b) Describe the role of the glass cockpit in modern engine monitoring.

(c) Identify the critical parameters monitored by the jet pipe temperature indicator.

12. Home Assignment and Follow Up	Prepare a brief note on the evolution of engine instruments from mechanical to glass cockpit systems, referencing LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose and operation of various safety devices used in electrical system of the aircraft. Safety precautions while working on electrical systems in aircraft.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min